Lesson notes · for you

Connected things, safely

IoT · Level 1 · Lesson 4

Time
45 min

Materials
Paper and a pencil for your table; An adult at home, and the devices' boxes or manuals if you have them; A phone, to check one app's permissions

Learning goals

- Explain what makes a thing part of the Internet of Things: a sensor, a small computer and a connection.
- Give examples at home, at school and in town.
- Protect yourself: permissions, default passwords, and what you share.

Activity: Connected-things check

1. Do this with a parent or guardian. Walk round your home and list everything that connects: phones, the Wi-Fi router, a TV, a camera, a watch, a solar kit.
2. In your table, write what each one senses (if anything) and how it connects: Wi-Fi, Bluetooth, the mobile network or a cable.
3. Ask the adult whether each factory password was ever changed. Write yes, no or don't know, never the password itself.
4. On one phone, open Settings, then Apps, then one app, then Permissions.
5. Together, find one permission the app doesn't need for its job, and switch it off. Write which one and why.

Only your own family's devices, and only with an adult. Never write real passwords on your sheet, and never try to get into someone else's device or account.

Discussion

- Which connected thing at school would be most useful, and what could go wrong?
- Why do attackers want cheap devices like cameras and routers?
- What should a device never collect about children?

Check yourself

- Name the three parts of an IoT device, and one way to keep it safe.

Try it this week
With an adult, list the connected things at home, check which factory passwords were changed, and switch off one app permission that isn't needed.

Worksheet

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Name _____ Class _____ Date _____

1 **A free puzzle game asks for your microphone, your contacts and your location. What do you allow?**

- ◯ All three, or the game won't install or run
- ◯ None: a puzzle needs none of them
- ◯ Only the location: that one is harmless

2 **Which of these are part of the Internet of Things?**

- ☐ A solar calculator
- ☐ A tank sensor that texts the caretaker when the water is low
- ☐ A pay-as-you-go solar kit that reports to its company
- ☐ A wind-up wall clock
- ☐ A fitness watch that sends your steps to a phone app

3 **Nobody would bother attacking a cheap camera in a small shop, so its factory password is safe enough.**

- ◯ True
- ◯ False

4 **Put in order how a farm soil reading reaches the farmer.**

- ☐ The farmer's phone shows an alert
- ☐ A computer far away receives and stores it
- ☐ The connection sends it over the mobile network
- ☐ The small computer turns the reading into a short message
- ☐ The soil sensor measures that the ground is dry

5 **Match each risk to the best defence.**

A camera still using the password admin	• • • •	Switch that feature off
A known weak spot in a router's software	• • • •	Switch off location in the camera app
A torch app asking for your contacts	• • • •	Tap Don't allow
Photos that carry where they were taken	• • • •	Install the update
A camera microphone you never use	• • • •	Change it to a long new password

Which connected thing in your home or school worries you most, and what one change would make it safer?